Reference-crystal selection for Crys-JEPA

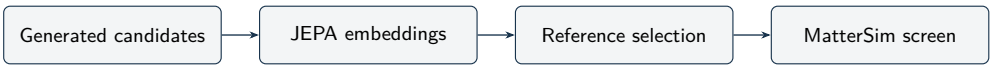
Group 2 Project Progress Report

October 30, 2026

Reports of the completed baselines and GNN evaluation.

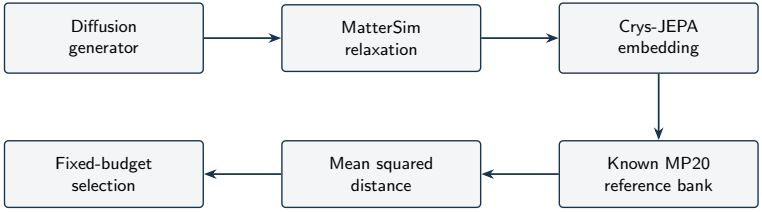
Motivation

- Diffusion models generate many candidate crystals.
- Crys-JEPA ranks candidates by their similarity to known reference crystals.
- The reference choice controls which candidates enter the expensive refinement loop.
- A useful reference set should capture structural context and improve the number of stable candidates selected at a fixed budget.



Crys-JEPA uses a known-crystal reference bank and a fixed candidate-selection budget.

Existing Crys-JEPA pipeline



- References come from 27,136 known MP20 training crystals.
- Generated candidates are used as queries to pick the top K references.
- A candidate passes the current screen when corrected MatterSim energy above the fixed hull is at most 0.1 eV/atom.

MatterSim is a comparative ML potential screen. It is not a DFT confirmation.

Original reference score

For a candidate c, collect known references whose elements are contained in the candidate system:

$$R(c) = \{r \in \mathcal{D}_{\mathrm{MP20}} : \mathrm{elements}(r) \subseteq \mathrm{elements}(c)\}.$$

The original score is

$$s(c) = \frac{1}{|R(c)|} \sum_{r \in R(c)} \|h_c - h_r\|_2^2.$$

- Smaller scores rank first.
- Every reference structure receives equal weight.
- Different polymorphs of one formula remain separate structures.
- We define: A chemical system is a set of elements. A reference subsystem is a known crystal whose elements are a subset of that system.

Why a chemical subsystem alone may miss useful structure

- A polymorph is a different crystal structure with the same formula.
- For Na, the MP20 bank contains several space groups and different energies.
- A formula-only view groups these structures by chemistry, but it does not describe their structural differences.
- A candidate may be close to a useful structure even when its best chemical match is outside the exact subsystem.

Material	Formula	E_{hull}	Space group
mp-1186040	Na	0.0179	223
mp-567772	Na	0.0130	220
mp-1186055	Na	0.0248	213
mp-973198	Na	0.0079	166
mp-127	Na	0.0123	229

Example from the MP20 polymorph inspection shared by the group.

Same chemistry does not mean the same structure or the same energy.

Analysis on crystal symmetry

Crys-JEPA embeddings of the 27,136 MP-20 crystals, compared against their space-group labels.

- **Communities align with symmetry.** Space-group AMI = **0.148** (0 = chance) — higher than the chemical-system AMI of 0.067; same under Euclidean and cosine graphs.
- **Directly decodable.** A linear probe predicts space group at **34.0%** accuracy, vs. 14.3% for guessing the most common group — well above chance for a many-class label.
- **Most groups concentrate.** **61 of 73** space groups cluster into a few communities far more than chance; rock-salt (SG 225) peaks at **4.15**×, then SG 216 (3.19×) and SG 187 (2.90×).

Inference: the embeddings carry a clear, non-random symmetry signal, consistent across all three tests.

Comparing the candidate selection from these methods

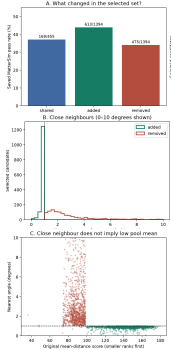
Shared set Candidates selected by both the original Crys-JEPA rule and the new rule. These candidates do not explain the difference.

Added set Candidates selected by the new rule but missed by the original rule. These are the extra candidates we study.

Removed set Candidates selected by the original rule but missed by the new rule. These are the candidates traded away.

Group	Count	MatterSim pass rate
Shared	455	37%
Added	1,394	44%
Removed	1,394	34%

Observed selection change



Panel A: selected-set change. The new rule adds 613 candidates and removes 475 relative to the shared set.

Panel B: nearest-reference angle. Added candidates have a close known reference, often inside the 1 degree gate.

Panel C: local versus pool score. A close neighbour does not always give a low mean distance to the full chemical pool. The plot shows why a local-neighbour rule can select different candidates from the original mean-distance rule.

Experiments

Baseline methods

Baseline	Definition
Original Crys-JEPA	Mean squared JEPA distance over eligible known MP20 references.
Cosine KNN, $k = 5$	Use the five closest known references by cosine distance.
Intersection KNN	Use the overlap between the original pool and cosine neighbours
Formula-balanced mean	Reduce the effect of formulas with many polymorphs.
Random references	Use a matched random reference pool as a negative control.

Baseline results at the 5% budget

Method	Passes	Yield	Δ vs original
Original Crys-JEPA	721	40.8%	–
Cosine KNN	722	40.9%	+1 (+0.1 pp)
Intersection KNN	760	43.0%	+39 (+2.2 pp)
Formula prior	693	39.2%	-28 (-1.6 pp)
Random references	679	38.4%	-42 (-2.4 pp)

- Intersection KNN is the strongest simple baseline.
- It combines chemical eligibility with the five nearest JEPA references.
- We rank candidates first, then evaluate the selected candidates with MatterSim.

MatterSim pass means corrected energy above the fixed hull is at most 0.1 eV/atom.

Graph model setup

Nodes

- Each node is one known MP20 reference crystal.
- The node stores its frozen 512-D JEPA embedding.
- Composition fractions describe the elements in the crystal.

Edges

- Connect each reference to its ten nearest references by cosine distance.
- Add the reverse edge so the graph is symmetric.
- Store cosine distance, raw squared distance, and chemical eligibility.

Prediction

- A generated candidate enters as a query, never as a graph node.
- Message passing updates each known reference using its neighbours.
- A candidate-conditioned readout scores the reference set and predicts a MatterSim pass.

The KNN graph defines who can exchange information. The edge features and candidate-conditioned readout decide how that information affects the final score.

Models compared

Model	Message-passing rule
GCN	Mix each reference with its neighbouring reference states using degree normalization, then update the node representation.
GraphSAGE	Compute a mean summary of neighbouring reference states and combine it with the current reference state.
GAT	Learn a separate importance weight for each neighbour before updating the reference representation.

Results across selection budgets

Method	1%	2%	5%	7%
Original Crys-JEPA	189	328	721	954
Cosine KNN	175 (-14)	318 (-10)	722 (+1)	950 (-4)
Intersection KNN	186 (-3)	333 (+5)	760 (+39)	1,013 (+59)
Angle gate	193 (+4)	348 (+20)	769 (+48)	1,042 (+88)
GCN	188 (-1)	333 (+5)	**774** (+53)	**1,067** (+113)
GAT	175 (-14)	329 (+1)	745 (+24)	1,022 (+68)
GraphSAGE	183 (-6)	325 (-3)	745 (+24)	1,018 (+64)

- At 5%, GCN selects 774 MatterSim passes versus 721 for Crys-JEPA.
- At 7%, GCN selects 1,067 passes versus 954 for Crys-JEPA.
- The table shows the best observed run for each model family at each budget. Seed details stay in the raw result files.

Counts are unique selected candidates passing the MatterSim hull screen.

Reference Set Comparison using Atomistic Models

- We need a proxy for comparing screening rules when new DFT calculations are outside the project scope.
- Crys-JEPA also compares embedding screening with atomistic energy models: **eSEN and CHGNet**.
- We use open **MEGNet and eSEN** weights to judge the same candidate selections under two additional models.
- Question: does a chemical reference pool, embedding kNN, or their union retrieve more proxy near-stable crystals at the same budget?

Change the screening references. Keep the candidate pool and evaluation hull fixed.

Crys-JEPA, Secs. 3 and 5, pp. 4 and 9: MatterSim evaluation and eSEN/CHGNet energy-based screening. MEGNet is our additional proxy.

Methodology: reference pools and equal screening budgets

- **Candidates: 6,258** MatterSim-relaxed, valid/unique/novel crystals from released batches 20–29.
- **References: 27,136** MP-20 crystals. Use the same frozen JEPA weights and reconstructed scaler for every embedding.

Pool	Reference selection rule
Chemical baseline	All MP-20 references within the candidate's chemical subsystems.
kNN	Closest $q \in \{5, 10, 15, 25\}$ raw JEPA embeddings by squared Euclidean distance.
Union	Chemical pool plus kNN, deduplicated by reference ID.

$$s_R(c) = \frac{1}{|R(c)|} \sum_{r \in R(c)} \|h_c - h_r\|_2^2$$

Rank the **combined ten-folder pool**. Top **1%, 5%, 10%**: **62, 312, 625** unique selections per method, with a seeded random control.

Hugging Face: liun-online/Crys_JEPA, exp_logs/base/mp_20/eval. These figures retain the combined-ranking results.

Proxy stability: energy above a fixed reference hull

Open model	Predicted quantity used for candidates and reference phases
MEGNet	Formation energy per atom, from `MEGNet-Eform-MP-2018.6.1`.
eSEN	Total cell energy divided by atom count, from `esen_30m_mptrj.pt`.

- Predict **58,377 MP2023 reference structures** and build a separate fixed hull for each model. Reuse stored geometries without another relaxation.
- Find the lowest predicted energy of a reference-phase mixture that matches the candidate's composition.

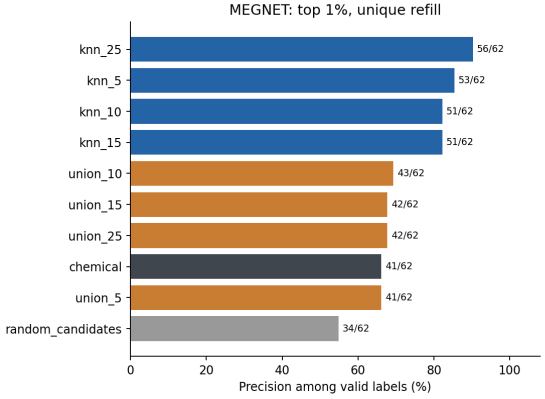
$$H_m(x_c) = \min_{\substack{\lambda_j \geq 0,\ \sum_j \lambda_j = 1 \\ \sum_j \lambda_j x_j = x_c}} \sum_j \lambda_j E_m(P_j), \qquad \Delta E_m(c) = E_m(c) - H_m(x_c).$$

Proxy near-stable: $\Delta E_m < 0.1$ **eV/atom.**

$p = 0.01$ means selecting the top **1%**. It is separate from the energy threshold. Union changes the JEPA score, not the hull bank.

MEGNet: materialyze/MEGNet-Eform-MP-2018.6.1. eSEN: facebook/OMAT24. Native model energies, no added MP corrections. Candidates never enter the fixed hull.

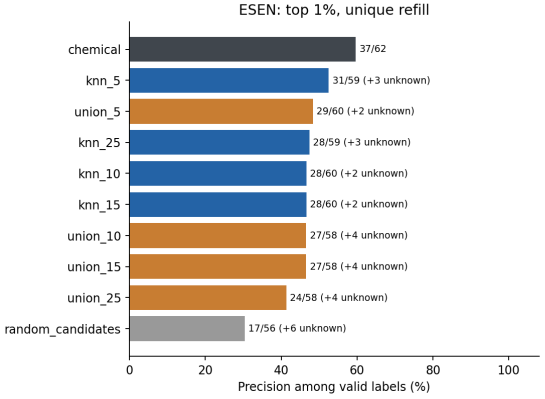
MEGNet results: kNN-25 leads at the 1% budget



- **kNN-25: 56/62 (90.3%)** versus chemical **41/62 (66.1%)**. **15 additional proxy passes.**
- Gain: **+24.2 percentage points** at the same selection budget.

Combined ranking of batches 20–29, top 1%: 62 selections. Fixed MEGNet hull distance < 0.1 eV/atom. All labels available.

eSEN results: lower pass rates and a baseline advantage



- **Chemical: 37/62 (59.7%)**; kNN-5: **31/59 (52.5%)**; kNN-25: **28/59 (47.5%)**.
- All methods have lower pass rates than with MEGNet. Both kNN sets have three unknowns.

Combined ranking, 62 slots. Precision uses valid labels; unknowns remain unknown. Pass rates do not measure model accuracy.

What the current results mean

- The angle gate and KNN controls change which chemical systems receive selection slots.
- Same-chemistry controls remove most of the earlier global gain.
- Same-formula pair accuracy is close to 50%, so the current score does not strongly order polymorphs by stability.
- The completed cluster evaluation shows a positive graph-model result at the fixed 5% budget.
- The atomistic-model comparison favours kNN under MEGNet, but eSEN does not corroborate that gain at 1%.
- The result is a MatterSim screening comparison. It does not prove lower true formation energy.

GNN with edge features. We added embedding-similarity, shared-space-group and electronegativity edge features to the reference graph; in terms of MatterSim screening yield at the fixed budget, they added nothing over the plain-weight baseline.

Current evidence and limits

- The reference bank is known MP20 training data.
- MatterSim labels provide a consistent comparative screen.
- MEGNet and eSEN disagree on which reference rule works best, so conclusions remain specific to the proxy.
- The current gains are screening-yield gains, not proof of lower true formation energy.
- We still need to analyse the results of individual success/reject cases and understand at a deeper level.

Reference quality must be tested by downstream candidate.